

Answer Key — Contact Forces: Muscular Force and Friction

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1 **(b) contact force** — contact forces need physical touch.
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- Q2 **(b) opposite to the direction of motion** — friction opposes the motion of the object.
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- Q3 **(b) rough surface** — rougher surfaces have more friction.
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- Q4 **(b) muscular force** — these use the muscular force of our body.
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- Q5 **(a) Both true, R explains A** — friction needs two touching surfaces, so it is a contact force.
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- Q6 **irregularities** — tiny surface irregularities cause friction.
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- Q7 **streamlined** — streamlined shapes reduce fluid friction.
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Short answer

- Q8 A force that acts only when objects are in physical contact; examples: muscular force and friction.
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- Q9 The force that acts when an object moves or tries to move over another surface; it acts opposite to the direction of motion.
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- Q10 On the nature of the surfaces in contact; friction is greater on rough surfaces and less on smooth ones.
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Long / application

- Q11 Friction opposes the box's motion and stops it. Different surfaces have different roughness, so friction differs: it is least on smooth glass (box slides farthest) and most on rough sand (box stops soonest).
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- Q12 It arises from irregularities of two surfaces in contact, so it needs touch — a contact force. Helpful: lets us walk/grip without slipping. Nuisance: slows moving vehicles and wears out parts (any reasonable examples).
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